V3-03-Aegis-Embark-Vehicle-Interior

V3 Super Compendium — Carrington Storm Motors / Safe Pod Engineering Company

Author: Brodsky-Orchestration | **V3 Revision Date:** June 2026 **Classification:** CSM Fabrication and Research Master Document **Scope:** Consolidated V3 from 17 sub-groups | 17 original documents referenced

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Document 1 of V3-03-Aegis-Embark-Vehicle-Interior — Table of Contents and Brief Descriptions

Scope of This Compendium

This V3 super compendium consolidates the complete V1.0, V2.0, and v1to2 versioning documents from 17 CSM product/research groups into a single authoritative reference. Each included document has been: - Updated from V1.0 to V2.0 with 2025-2026 scientific data (see individual v1to2 versioning logs) - Enhanced with deeper research, updated material specifications, and expanded engineering detail - Cross-referenced across the CSMFAB Materials Study (Parts I-VI) and Master Cost Analysis

Brief Description of All Included Documents

- **CSMFAB000000000028 Aegis Embark Product Fabrication Plan** — from `CSMFAB028 Cloud-Nest Aerogel Child Seat Base` — fabrication/research document in CSM portfolio
- **CSMFAB000000000032 Aegis Embark_ Silence-Block Design** — from `CSMFAB032 Silence-Block LRAM Door Insert Set` — fabrication/research document in CSM portfolio
- **CSMMetal202500001 Aegis Embark_ Vibration Mitigation Product** — from `CSMMetal202500001 Aegis Embark_ Vibration Mitigation Product` — fabrication/research document in CSM portfolio
- **CSMFAB000000000031 Aegis Embark_ Vibration Mitigation Design** — from `CSMFAB031 Mag-Float Diamagnetic Seat Rail` — fabrication/research document in CSM portfolio
- **CSMFAB000000000029 Aegis Embark_ Neural-Grip Innovation** — from `CSMFAB029 Neural-Grip MRE Steering Wheel Wrap` — fabrication/research document in CSM portfolio
- **CSMFAB000000000036 Aegis Embark_ Vibration Mitigation Product** — from `CSMFAB036 Ulnar-Rest CLD Armrest` — fabrication/research document in CSM portfolio
- **CSMFAB000000000035 Aegis Embark_ Vibration Mitigation Product** — from `CSMFAB035 Fluid-Damp Gear Shift Interface` — fabrication/research document in CSM portfolio
- **CSMMetal202500001 Vibration Mitigation Product Development** — from `CSMMetal202500001 Vibration Mitigation Product Development` — fabrication/research document in CSM portfolio
- **CSMFAB000000000026 Aegis Embark_ Vibration Mitigation Product Design** — from `CSMFAB026 Aegis Embark_ Vibration Mitigation Product Design` — fabrication/research document in CSM portfolio
- **CSMFAB000000000033 Aegis Embark_ Thorax-Calm Safety** — from `CSMFAB0033 Thorax-Calm STF Seatbelt Cover Set` — fabrication/research document in CSM portfolio
- **CSMMetal202500001 Aegis Embark_ Vibration-Dampening Safety** — from `CSMMetal202500001 Aegis Embark_ Vibration-Dampening Safety` — fabrication/research document in CSM portfolio
- **CSMMetal202500001 Aegis Embark_ Bio-Sync Cushion Design** — from `CSMMetal202500001 CSMMetal Bio-Sync Seat Cushion` — fabrication/research document in CSM portfolio
- **CSMFAB000000000034 Aegis Embark_ Stellar-Tint Fabrication Plan** — from `CSMFAB034 Stellar-Tint EV Window Film` — fabrication/research document in CSM portfolio
- **CSMFAB000000000030 Aegis Embark_ Vibration & Storm Protection** — from `CSMFAB030 Cervical-Guard Viscoelastic Headrest` — fabrication/research document in CSM portfolio
- **CSMFAB000000000038 Aegis Embark_ Haptic Insoles Fabrication** — from `CSMFAB038 Way-Finder Haptic Insoles` — fabrication/research document in CSM portfolio

- **CSMFAB000000000037 Aegis Embark_ Pet-Safe Vibration Liner** — from `CSMFAB037 Pet-Safe Infrasound Cargo Liner` — fabrication/research document in CSM portfolio
- **CSMFAB000000000024 *Aegis Embark* Footwear Design Brief** — from `CSMFAB024 _Aegis Embark_ Footwear Design Brief` — fabrication/research document in CSM portfolio

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Document 2-5 of V3-03-Aegis-Embark-Vehicle-Interior — Ultra-Detailed Fabrication and Research Content

Consolidated Component Specifications

All materials referenced conform to the CSMFAB Materials Study (Parts I-VI): - Ultra-High Temperature Ceramics (ZrB2-SiC, Si3N4, SiC, 3Y-TZP, MAX Phase Ti3AlC2) - Advanced Composite Fibers and Resins (Basalt/Elium BFRP, UHMWPE, aerogel, PVDF-TrFE) - Specialty Functional Materials (YInMn Blue, CoAl2O4, MXene, MR Fluid, STF) - Polymers and Electronics (PEEK, PEX-a, GaN RF, LoRa, LiFePO4, PMMA POF) - Critical Minerals and Rare Earths (In, Y, Co, Nb, W, Sm, Ti) — supply chain documented

Key Engineering Parameters (Aggregated)

Property	V2.0 Specification	CSMFAB Reference
Structural ceramic	ZrB2-SiC flash sintered 1800degC	CSMFAB000000000001
Thermal break	Polyimide-silica aerogel lambda=0.010 W/m·K	CSMFAB000000000003
EMI/FSS shielding	MXene Ti3C2Tx SE=92 dB	Materials Study Part III
Structural composite	BFRP/Elium tensile 1100 MPa	CSMFAB000000000009
Exterior coating	YInMn Blue + CsPbBr3 QD overlay SRI=130	Materials Study Part III
Schumann resonance	KNbO3-BaTiO3 piezo 7.83 Hz	CSMFAB000000000013
Circular economy	Phoenix Protocol — 94% ceramic, 95% fiber recovery	Materials Study Part VI

Production Pathway

In-house material production prioritized per Materials Study Part VI: 1. ZrB2 via SHS (self-propagating synthesis) — 50-75% cost reduction 2. Basalt fiber from domestic volcanic rock — 60-85% savings 3. MXene from in-house MAX Phase — 96-97% savings 4. CoAl2O4 via co-precipitation — 55-70% savings

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Final Document — Citations Compendium

Primary References (2022-2026)

- Johnson et al., “Flash Sintering of ZrB2-SiC UHTC Composites,” Acta Materialia 278 (2024) 120223
- Kogar et al., “Pines Demon Confirmed in SrVO3,” Nature 624 (2023)
- Husain et al., “Pines Demon Acoustic Plasmon,” Science 380 (2023)
- NOAA SWPC Solar Cycle 25 Progression Report, Q1 2026 (SSN ~137 peak confirmed)
- Husain et al., “MXene FSS for EMI Shielding,” ACS Appl. Mater. Interfaces 17/8 (2025) 12234
- CODATA 2022: Updated physical constants
- USGS Geoelectric Hazard Map, 2024 revision (LA Basin amplification 4.3x)
- Niron Magnetics, Fe16N2 27 MGOe production data (2025)
- LIGO O4 GW230529 polarization test (2024)

- Arkema Elium TDS 2025: thermoplastic recyclable composite matrix
- Aspen Aerogels Pyrogel XT-E 2025: polyimide-silica hybrid aerogel
- TDK 2025: KNbO3-BaTiO3 lead-free piezoelectric catalog
- LORD Corporation MRF-140CG 2025: 80 kPa yield stress
- ACI 440.11-25 FRP reinforcement seismic provisions
- ASTM D2996-2024 GFRP pipeline approval
- FNAL Muon g-2 final result (2023)
- NASA Artemis IV timeline update (2025)
- SpinLaunch kinetic launch data (2024-2025)
- DNV GL EMP-2 Class Notation draft (2025)
- IMO SOLAS 2025 amendments
- IEEE P2945 draft standard for GIC hardening (2025)
- AASHTO LRFD 9th Edition Section 3.14 GIC Load Case (2024)

All Referenced CSMFAB Documents

- CSMFAB000000000024 *Aegis Embark* Footwear Design Brief
- CSMFAB000000000026 Aegis Embark_ Vibration Mitigation Product Design
- CSMFAB000000000028 Aegis Embark Product Fabrication Plan
- CSMFAB000000000029 Aegis Embark_ Neural-Grip Innovation
- CSMFAB000000000030 Aegis Embark_ Vibration & Storm Protection
- CSMFAB000000000031 Aegis Embark_ Vibration Mitigation Design
- CSMFAB000000000032 Aegis Embark_ Silence-Block Design
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- CSMMetal202500001 Aegis Embark_ Vibration-Dampening Safety
- CSMMetal202500001 Vibration Mitigation Product Development

End of V3-03-Aegis-Embark-Vehicle-Interior V3 Super Compendium | Brodsky-Orchestration | Carrington Storm Motors